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A systematic review and meta-analysis on the benefits of resistance exercise in patients with fibromyalgia

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ABSTRACT

Objectives: This systematic review and meta-analysis aimed to analyse the effects of resistance-based exercise programs on pain, fatigue, depression, anxiety, sleep quality, quality of life, physical function, and muscle strength in patients with fibromyalgia. Additionally, we investigated whether these effects were influenced by demographic, clinical, or exercise prescription characteristics.

Methods: A systematic search in seven databases was conducted up to February 2024. Eligible randomized controlled trials examined the effects of resistance-based exercise programs on outcomes of interest in patients with fibromyalgia. Outcomes were analysed using a three-level mixed-effects meta-analysis and meta-regression, with statistical significance set at an α level of 0.05.

Results: Twenty-two studies involving 1235 participants were included. Resistance-based exercise showed small but significant improvements in pain (−0.40 SMD, 95%CI: −0.55 to −0.25), fatigue (−0.48 SMD, 95%CI: −0.69 to −0.26), depression (−0.46 SMD, 95%CI: −0.70 to −0.22), and anxiety (−0.31 SMD, 95%CI: −0.43 to −0.19). Small gains were observed also observed in sleep quality and physical function, while lower-limb muscle strength demonstrated a large effect size (1.17 SMD, 95% CI: 0.89 to 1.44). Subgroup analyses indicated benefits irrespective of age, BMI, or duration of illness. Adding aerobic or flexibility exercises did not result in additional effects.

Conclusions: Resistance-based exercise programs provide modest but meaningful benefits for managing multiple symptoms of fibromyalgia. A low-dose, moderate-intensity protocol may be sufficient to yield improvements and should be recommended as part of a multidisciplinary treatment approach.

1. Introduction

Fibromyalgia is a chronic disorder affecting up to 5.0% of the global population (Neumann and Buskila, 2003; Queiroz, 2013). This disorder is characterized by widespread pain and tenderness in muscles and soft tissue (Neumann and Buskila, 2003; Queiroz, 2013). The mechanism

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responsible for this remains a topic of intense debate in the literature, with nociceptive, neuropathic and more recently, nociplastic pain proposed as the primary pain source (Fitzcharles et al., 2021; Martínez-Lavín, 2022), reflecting the progressive evolution in the understanding of fibromyalgia (Clauw, 2014; Littlejohn and Guymer, 2020; Wolfe and Walitt, 2013). Beyond chronic pain, patients are likely also to experience increasing fatigue, poor sleep quality, depression, and anxiety, which have a substantial impact on activities of daily living, quality of life and well-being. The direct and indirect costs of treatment, medication and necessary lifestyle adjustments can reach ~$36,000 per year in the US (D'Onghia et al., 2022). Standard fibromyalgia treatments include anti-inflammatories, analgesics, and muscle relaxants (Aster et al., 2022), as well as antidepressants and anxiolytics (Gracely et al., 2012; Yepez et al., 2022). Despite the variety of medications, managing symptoms and coping with chronic pain and fatigue is difficult, often requiring adjustments in lifestyle and self-care routines.

Over time, the management of fibromyalgia has evolved from a predominantly passive approach (e.g., rest and symptom avoidance) to strategies that increasingly emphasise active rehabilitation, particularly through structured exercise interventions (Majdoub et al., 2023; Neelapala et al., 2023). According to the latest *European League Against Rheumatism* (EULAR) recommendations (Macfarlane et al., 2017), exercise is considered one of the most effective therapies in managing fibromyalgia. Although aerobic and flexibility activities are commonly prescribed in this population, resistance-based exercise (i.e., resistance training alone or combined with other activities) has been increasingly investigated in recent years due to its distinct physiological and clinical effects (Guachizaca Moreno et al., 2025; Larsson et al., 2015). Resistance exercise promotes neuromuscular adaptations that enhance physical function and may contribute to pain modulation, differing from aerobic and flexibility exercise, which are primarily associated with cardiorespiratory fitness and mobility (Bidonde et al., 2017; Busch et al., 2013; Kim et al., 2019). This aligns with the increasingly recognition of fibromyalgia as a multidimensional condition, in which outcomes extend beyond pain to include fatigue, psychological well-being, quality of life, and physical function (Arnold et al., 2008; Fitzcharles and Yunus, 2012). Specifically, two Cochrane reviews (Busch et al., 2007, 2013) have identified moderate pain reduction and improvements in physical function following resistance exercise programs.

Nevertheless, from a therapeutic development perspective, the specific circumstances in which resistance exercise results in clinically meaningful benefits in patients with fibromyalgia are yet to be determined. First, it is important to determine the extent of resistance exercise effects in patients with fibromyalgia. Indeed, previous meta-analyses (Couto et al., 2022; Han et al., 2024; Niu et al., 2024; Rodríguez-Almagro et al., 2023) had reported large effects [>1.3 standardised mean difference (SMD)] on pain and other outcomes after a resistance exercise program. However, such results are not observed in other clinical populations (i.e., cancer, low back pain, hip/knee osteoarthritis (Fleckenstein et al., 2022; Lopez et al., 2021; Plinsinga et al., 2023; Sasaki et al., 2022), requiring a more comprehensive assessment of the literature. Second, it is unclear whether demographic and clinical characteristics such as age, body mass index (BMI), duration of illness and overall symptomatology may affect responses to resistance exercise programs. Third, further research is necessary to determine if combining other exercise modalities with resistance exercise programs or prescribing higher doses and intensities of resistance exercise will result in better effects on pain, fatigue, depression, anxiety, sleep quality, quality of life, physical function and muscle strength. Such information could substantially improve the management of fibromyalgia (Macfarlane et al., 2017) by identifying when and for whom exercise provides clinically meaningful benefits, supporting the development of personalized and more effective exercise interventions. These could help in tailoring treatments for patients with higher care needs (Macfarlane et al., 2017).

In this systematic review and meta-analysis, we aimed to determine the effects of resistance-based exercise programs on pain, fatigue, depression, anxiety, sleep quality, quality of life, physical function and muscle strength in patients with fibromyalgia. Additionally, we examined whether these effects were modified by demographic and clinical characteristics (i.e., age, body mass index, duration of illness and overall symptomatology) and exercise prescription characteristics (i.e., exercise components, duration, weekly volume and intensity). These findings may help refine the role of resistance-based exercise within fibromyalgia management, with potential implications for future guidelines and the selection of more effective exercise strategies in clinical practice.

2. Methods

All procedures undertaken in the present study were reported in accordance with the Cochrane Back Review Group (CBRG) guidelines (Furlan et al., 2009), the Implementing Prisma in Exercise, Rehabilitation, Sport Medicine and Sports Science (PERSiST) (Ardern et al., 2022) and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Liberati et al., 2009; Page et al., 2020). This review was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under the identifier CRD42023495434.

2.1. Eligibility criteria

Studies were included in this systematic review if meeting the following criteria regarding participants, intervention, comparator, outcomes and study design (PICOS): i) studies involving patients with fibromyalgia aged 18 years or older; ii) studies implementing resistance-based exercise programs (i.e., interventions including resistance exercise as one of the components) delivered as supervised or unsupervised programs (i.e., unsupervised home-based programs); iii) studies implementing a non-exposed control group, such as those with no formal intervention, waitlist controls, or low-intensity exercise programs that constitute minimal exposure for the outcomes of interest (e.g., stretching, relaxation, Tai chi, walking exercise group) as comparators; iv) studies measuring pain, depression, anxiety, sleep quality, fatigue and quality of life using validated questionnaires or scales, physical function assessed by a 6-min walking test, and lower-limb muscle strength measured by a one-repetition maximum test (1-RM); and v) randomised controlled trials. Fatigue was not originally planned for inclusion in this systematic review and meta-analysis, and this deviation from the original protocol submitted to PROSPERO is characterised as *addition*. This outcome was included given its high prevalence and relevance for patients with fibromyalgia (Vincent et al., 2013).

The exclusion criteria were: i) studies with interventions lasting less than 4 weeks; ii) studies involving resistance exercise interventions combined with any nutritional approach (e.g., healthy diet, caloric restriction, protein supplementation); iii) studies not reporting sufficient information or specific outcomes included in this review; and iv) studies published in a language other than English, Portuguese or Spanish.

2.2. Search strategy and study selection procedure

A systematic search was conducted by a researcher (PL) using *CINAHL*, *Embase*, *LILACS*, *PubMed*, *Scielo*, *SPORTDiscus* and *Web of Science* databases from inception to February 2024. The search strategy was undertaken using controlled vocabulary and free-text terms as presented in the Supplementary Appendix S1. a manual search of references in selected studies (i.e., backward citation chasing) was performed to detect potentially eligible articles for inclusion in addition to database searching. Titles and abstracts were first independently evaluated following the eligibility criteria during the screening phase. Eligibility was assessed independently in duplicate (EC and TAW), with differences resolved by consensus, or by a third reviewer (PL) when disagreements occurred. In case of abstracts that did not provide sufficient information, they were selected for full-text evaluation. Full-text articles meeting criteria were retrieved and read independently by both reviewers and assessed for inclusion in the study.

2.3. Data extraction

Data extraction was performed in duplicate (EC and TSW) and checked by a third researcher (PL) via a standardised form. Study information including sample size, age, baseline BMI, number of female participants, duration of illness, overall symptomatology (assessed by the Fibromyalgia Impact Questionnaire [FIQ]), number and type of medications, experimental design, and exercise program (i.e., intervention duration, resistance exercise volume, resistance exercise intensity, aerobic exercise volume, aerobic exercise intensity), and retention (i.e., number of participants that completed the study) and attendance (i.e., percentage of sessions attended) were extracted along with the outcomes of interest. Outcomes were assessed at baseline and post-intervention, with within- and between-group mean difference extracted in their absolute units and for the longest period of the intervention. Studies that did not provide dispersion values of change, such as standard deviation (SD), standard errors or 95% confidence intervals (95% CI), had the SD of the change calculated assuming a correlation of r = 0.5 between the baseline and post-intervention assessment measures by the square root of $\left(\left(SD^2_{Baseline} + SD^2_{Post-intervention}\right) - \left(2 \times r \times SD_{Baseline} \times SD_{Post-intervention}\right)\right)$ (Lefebvre et al., 2011). For studies containing multiple intervention arms versus control groups, only data from those comprising resistance exercise as part of the intervention were extracted. When graphs were used instead of numerical data, the graphs were measured through their plots using a specific tool for data extraction (WebPlotDigitizer, San Francisco, CA) (Drevon et al., 2017).

2.4. Study risk of bias assessment and certainty of evidence (GRADE)

The risk of bias was evaluated according to the 2nd version of the Cochrane risk-of-bias tool for randomised trials (RoB 2) (Sterne et al., 2019), with each assessment focused on the outcome level. The six-domain instrument includes: (1) randomisation process; (2) deviation from intended interventions; (3) missing outcome data; (4) measurement of the outcome; (5) selection of the reported result; and (6) overall bias. Overall risk of bias was expressed as "low risk of bias" if all domains were classified as low risk, "some concerns" if some concern was raised in at least one domain but not classified as at high risk in any other, or "high risk of bias" if at least one domain was classified as high risk, or two domains or more had some concerns (Sterne et al., 2019). The study risk of bias assessment for all included studies was performed independently by two reviewers (CTM and PC), with disagreements resolved by consensus. In addition, the certainty of evidence was assessed using the *Grading of Recommendations Assessment, Development and Evaluation* (GRADE) approach. The several domains considered for downgrading the certainty of evidence were risk of bias, imprecision, inconsistency, indirectness and publication bias (i.e., based on comparison-adjusted funnel plot).

2.5. Data analysis

A robust variance estimation approach was undertaken to account for the nested structure of the effect sizes calculated from the studies included (i.e., effects nested within categories nested within studies) (Cheung, 2014). This approach was undertaken given the availability of several dependent outcomes from the same study. Therefore, a three-level mixed-effects meta-analysis with *study* included as a random effect was performed to examine the effect of the resistance-based exercise programs on the outcomes of interest. The pooled effect estimated from the outcomes were obtained and expressed as SMD and mean difference (MD) when the outcomes were consistently assessed by a specific questionnaire/scale and method. All values used for analyses are presented in Supplementary Table S1–S8. Cluster robust point estimates 95% confidence intervals (95% CI) were provided, weighted by inverse sampling variance to account for the within- and between-study variance (τ^2). Restricted maximal-likelihood estimation was used in all models. Statistical significance was assumed when the SMD or MD effect was below an α level of $P \leq 0.05$. According to Cohen (1992), SMD values of 0.0 to ≤0.5 represent small; 0.51 to 0.79, medium; and ≥0.8, large effects. Statistical heterogeneity was assessed using the Cochran Q test. A threshold P value of 0.1 and values greater than 50% in I^2 were considered indicative of high heterogeneity. Publication bias was explored by contour-enhanced funnel plots and Egger's test (Peters et al., 2008) when more than 10 studies were available.

Subgroup analyses were provided for i) assessment method; ii) fibromyalgia criteria; and iii) exercise modality. Multilevel models with robust estimates were produced for each subgroup, and fixed effects with moderator's model were used to compare the models. Meta-regression models were undertaken to test the association between potential demographic and clinical moderators as well as resistance exercise prescription characteristics with the outcomes of interest when more than 10 studies were available. Hypothesised demographic and clinical moderators included age (years), BMI ($kg.m^{-2}$), duration of illness (months) and FIQ (points). Hypothesised moderators of resistance exercise response included number of sessions measured by *duration* × *frequency*, weekly volume measured by *frequency* × *exercise* × *sets*, and resistance exercise peak intensity measured by the highest %1-RM prescribed throughout the program. Statistical significance was assumed when β ± SE was below an α level of $P \leq 0.05$. Analyses were conducted using the package *metafor* (Viechtbauer, 2010) and *club-Sandwich* (Pustejovsky and Tipton, 2018) from R (R Core Team, version 4.0.3., 2020).

3. Results

A total of 811 potential records were identified following deletion of duplicates and records marked as ineligible. Of these, 488 records were excluded based on titles and abstracts due to their irrelevance to the research question, resulting in 323 records deemed eligible for full-text review. A total of 295 reports were excluded due to reasons described in Fig. 1. After including one additional study (Valkeinen et al., 2004) via reference lists, a total of 27 articles (Andrade et al., 2022; Arakaki et al., 2021; Assumpção et al., 2018; Corrales et al., 2010; Ericsson et al., 2016; Gandhi et al., 2002; García-Martínez et al., 2012; Gavi et al., 2014; Giannotti et al., 2014; Häkkinen et al., 2001; Izquierdo-Alventosa et al., 2020a, 2020b; Jones et al., 2002; Kayo et al., 2012; Kingsley et al., 2005; Kolak et al., 2022; Larsson et al., 2015; Park et al., 2021; Rooks et al., 2007; Sañudo et al., 2010, 2011, 2012; Sauch Valmaña et al., 2020; Silva et al., 2019; Valkeinen et al., 2004, 2005, 2008) representing 22 studies were included in the present systematic review and meta-analysis.

3.1. Study, participant, and intervention characteristics

A total of 1235 patients with fibromyalgia (1224 females and 11 males) were included with a median age of 51.4 (interquartile range [IQR]: 47.4-54.0) years and BMI of 28.2 (IQR: 27.1-28.6) $kg.m^{-2}$ at study level. Most patients (77%) were diagnosed using the *American College of Rheumatology (1990) Criteria* and the median duration of fibromyalgia before the study commencement was 7.8 (IQR: 6.3-10.0) years. A total of 169 patients (13.7%) reported using anti-inflammatory, analgesic and/or muscle relaxers before the commencement of five studies. Regarding the overall symptomatology, the median FIQ score was 62.3 (IQR: 58.9-66.0) points after analysing the reporting of 16 studies, with two studies reporting average scores above 70.0 points, which is the threshold for severely affected patients. Median retention was 82.8% (IQR: 75.8%-90.5%), with a total of 972 patients completing the studies.

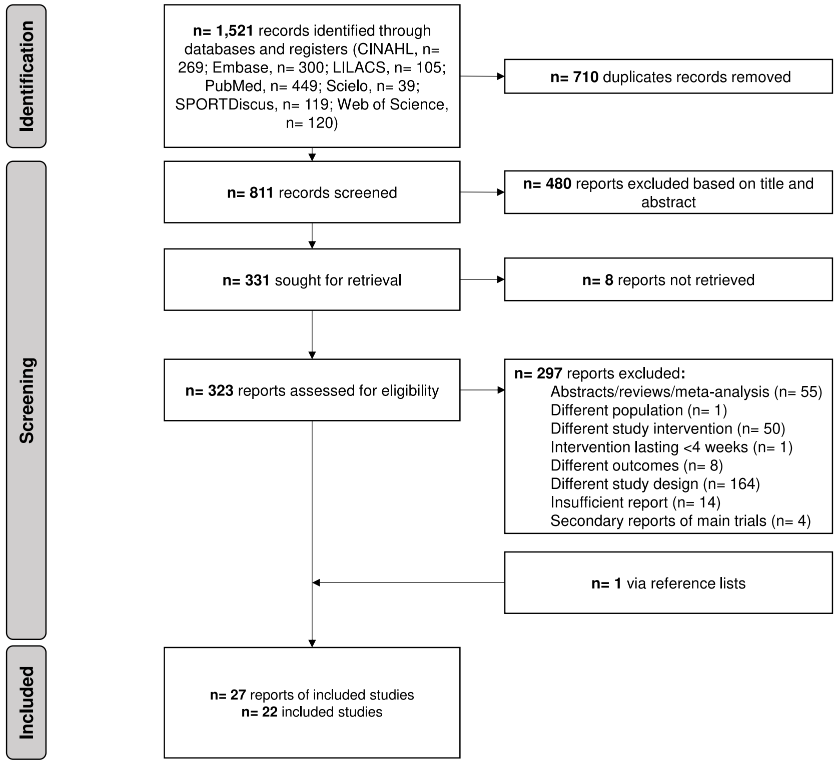

Fig. 1. Flow chart of study selection process.

A total of 22 intervention groups were analysed. Eleven (50.0%) interventions were resistance exercise only, followed by combined resistance, aerobic and flexibility interventions (6 studies, 27.3%), combined resistance and aerobic (3 studies, 13.6%), and combined resistance and flexibility (2 studies, 9.1%). All interventions were supervised. The median duration of the programs was 12 (IQR: 12-16) weeks, with a median of 31 (IQR: 24-41) sessions prescribed at study level. Regarding the resistance exercise prescription, 16 studies (72.7%) included reported the frequency, number of resistance exercises prescribed and number of sets per exercise, with a median volume of 23 (IQR: 18-72) sets per week for the whole body at study level. The peak intensity was reported by nine studies with a median of 80% (IQR: 50%-80%) of 1-RM. Aerobic exercise within combined resistance and aerobic exercise was prescribed in nine studies, with a median duration of 18 (IQR: 13-21) min per session at a peak intensity of 70% (IQR: 70%-70%) of maximum heart rate.

For patient-reported outcomes, 14 studies (63.6%) assessed pain, and 14 studies (63.6%) assessed fatigue, followed by 13 studies (59.1%) assessing general quality of life, 11 studies (50.0%) assessing depression, six studies (27.3%) assessing sleep quality and five studies (22.7%) assessing anxiety. Physical function measured by the 6-min walking test was examined in six studies (27.3%), while muscle strength measured by the 1-RM test was examined in five studies (22.7%). The characteristics of the individual studies are presented in Supplementary Table S9.

3.2. Risk of bias

High risk of bias was identified in 18 out of 21 studies (85.7%) examining pain, 11 out of 13 studies (84.6%) examining quality of life, 12 out of 14 studies (85.7%) examining fatigue, and eight out 11 studies (72.7%) examining depression. Five out of six studies (83.3%) and three out of five studies (60%) examining physical function and lower-limb muscle strength were deemed high risk of bias, respectively. All studies were deemed high risk of bias for anxiety (five studies) and sleep quality (five studies). The risk of bias assessment for each outcome is presented in Supplementary Table S10–S17.

3.3. Pain

Resistance-based exercise programs resulted in a small effect on pain reduction compared to control groups, with a SMD of −0.40 (95% CI: −0.55 to −0.25, $p < 0.001$) based on the analysis of 57 effect sizes across 22 studies involving a total of 995 patients (Fig. 2). In regard to the assessment method, a reduction of −11.48 points (95% CI: −15.96 to 6.99, $p < 0.001$) was observed using the visual analogue scale (0-100 mm) and an increase of 8.08 points (95% CI: 1.63 to 14.53, $p = 0.020$) using the *SF-36 bodily pain scale*. Subgroup analyses indicated that pain reduction results were relatively consistent (effect size range: −0.56 to −0.40 SMD) in studies utilising the *American College of Rheumatology (1990)* and *2016 Criteria*, but not the *2010 Criteria* (effect size: 0.18 SMD; Table 1). Moreover, relatively similar effects (effect size range: −0.46 to −0.38 SMD) were observed following resistance exercise, combined resistance and aerobic exercise and combined resistance, aerobic and flexibility exercise. However, reductions in pain were less likely to be meaningful following combined resistance and flexibility exercise (effect size: −0.10 SMD).

The heterogeneity I^2 was 0% and no presence of publication bias was detected ($\tau = -1.5$, $p = 0.142$; Supplementary Fig. S1). In the meta-regression, demographic and clinical characteristics including age, BMI, duration of illness, and overall symptomatology, were not associated with changes in pain ($p = 0.486$ to 0.773) following resistance-based exercise programs (Table 2). Additionally, characteristics of resistance exercise prescription such as duration, weekly volume and peak intensity, as well as the inclusion of aerobic and flexibility exercise were not significantly associated with changes in pain ($p = 0.350$ to 0.721). Based on the GRADE approach, the certainty of evidence was considered low.

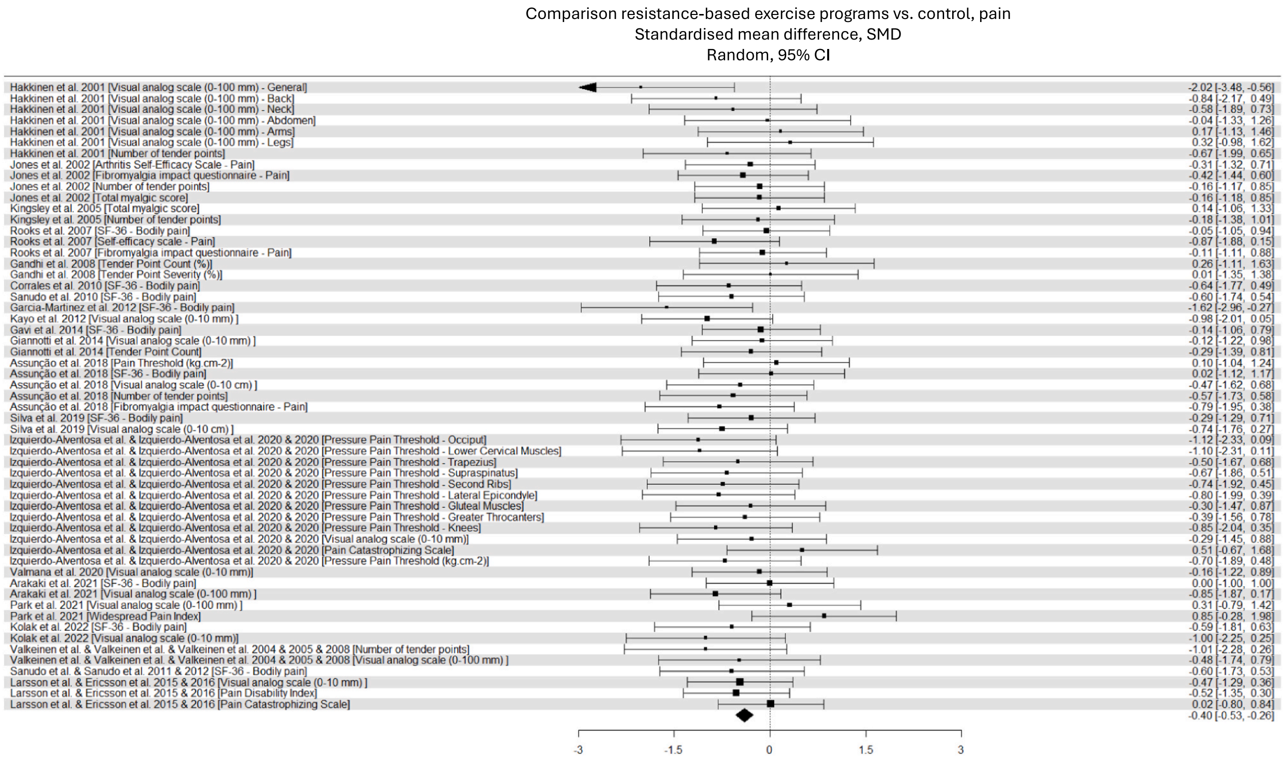

Fig. 2. Forest plot of effects of resistance-based exercise programs compared with control on pain in patients with fibromyalgia. Overall analysis conducted with three-level mixed-effects model. CI, confidence interval; SMD, standardised mean difference.

Table 1
Overall and subgroup analyses of resistance-based exercise effects on pain in patients with fibromyalgia.

Analysis	Descriptive			Random-effect meta-analysis		Heterogeneity			Certainty of evidence
	k	ES	n	Effect size (95% CI)	p-value	Q	I^2	p-value	
Overall, SMD	22	57	995	−0.40 (−0.55 to −0.25)	<0.001	37.0	0%	0.976	⊖⊖⊕⊕ Low[a,b]
Questionnaire									
Visual analog scale (0-100 mm), points	12	17	579	−11.48 (−15.96 to −6.99)	<0.001	334.3	100%	<0.001	
SF-36 – Bodily pain, points	10	10	463	8.08 (1.63 to 14.53)	0.020	117.0	49.8%	<0.001	
Fibromyalgia criteria									
1990, SMD	16	38	774	−0.40 (−0.51 to −0.29)	<0.001	22.5	0%	0.971	
2010, SMD	2	4	81	0.18 (−4.79 to 5.15)	0.720	2.4	31%	0.493	
2016, SMD	3	15	140	−0.56 (−1.31 to 0.18)	0.080	6.8	0%	0.942	
Exercise modality									
Resistance, SMD	10	29	557	−0.38 (−0.50 to −0.26)	<0.001	16.1	0%	0.964	
Resistance + Flexibility, SMD	2	4	68	−0.10 (−8.79 to 8.60)	0.910	5.8	68%	0.120	
Resistance + Aerobic, SMD	4	15	153	−0.43 (−1.29 to 0.42)	0.170	6.9	7%	0.937	
Resistance + Aerobic + Flexibility, SMD	6	9	217	−0.46 (−0.94 to 0.02)	0.060	6.3	0%	0.613	

95% CI, 95% confidence intervals; ES, number of effect sizes; I^2, percentage of variation across studies that is due to heterogeneity; k, number of studies; n, number of participants; Q, Cochran's Q test of heterogeneity.

[a] Certainty of evidence downgraded due to risk of bias, with studies most studies presenting with high risk in the risk of bias assessment.

[b] Certainty of evidence downgraded due to indirectness, with different results observed across the interventions and assessment methods examined.

3.4. Quality of life

Thirteen studies comprising a total of 564 patients were analysed in the main model for quality of life. Resistance-based exercise programs resulted in a small effect of −0.28 SMD (95% CI: 0.09 to 0.46, p = 0.010) compared to control groups (Supplementary Fig. S2). In terms of the assessment method, an improvement of 3.04 points (95% CI: 0.57 to 5.52, p = 0.020) was noted using the *SF-36 general health scale*. The main effect was primarily observed in studies utilising the *American College of Rheumatology (1990)* (0.26 SMD, 95% CI: 0.07 to 0.46, p = 0.010), and in those following resistance exercise and combined resistance, aerobic and flexibility exercise (effect size range: 0.23 to 0.29 SMD), although not statistically significant (p = 0.080 to 0.090; Table 3).

The heterogeneity I^2 for the main model was 0%. Contour-enhanced funnel plot indicated a presence of publication bias (τ = 3.9, p = 0.002; Supplementary Fig. S3). Meta-regression analyses indicated that neither age (p = 0.663) nor overall symptomatology (p = 0.702) were associated with changes in quality of life (Supplementary Table S18). Additionally, resistance exercise prescription characteristics such as duration and weekly volume, as well as the inclusion of aerobic and flexibility exercise, were not significantly associated with changes in quality of life (p = 0.116 to 0.897). A very low certainty of evidence was graded for quality of life based on the GRADE approach.

Table 2
Meta-regression on demographic, clinical and exercise prescription characteristics and changes in pain in patients with fibromyalgia..

Moderator	k	ES	n	β ± SE	p-value
Demographic and clinical characteristics					
Age, years	20	53	903	−0.01 ± 0.02	0.621
Body mass index at baseline, kg. m^{-2}	14	28	691	−0.06 ± 0.08	0.486
Duration of illness, month	10	24	502	0.00 ± 0.00	0.773
FIQ, point	16	32	716	−0.00 ± 0.01	0.556
Resistance exercise prescription characteristics					
Duration, weeks	22	57	995	−0.02 ± 0.02	0.350
Weekly volume, sets per week	16	42	817	−0.00 ± 0.00	0.430
Peak intensity, % of 1-RM	9	24	496	-	-
Other exercise components					
Aerobic, Yes vs. No	20	53	903	0.12 ± 0.14	0.395
Flexibility, Yes vs. No	20	53	903	−0.08 ± 0.21	0.721

β, meta-regression coefficient; 1-RM, one-repetition maximum; ES, number of effect sizes; k, number of studies; n, number of participants; SE, standard error..

3.5. Depression

Resistance-based exercise programs resulted in a small decrease in depression compared to control groups, with a SMD of −0.46 (95% CI: −0.70 to −0.22, $p < 0.001$) based on the analysis of 12 effect sizes from 11 studies, encompassing a total of 560 patients (Supplementary Fig. S4). A reduction of −4.58 points (95% CI: −8.08 to −1.08, $p = 0.020$) was observed in the *Beck Depression Scale*. Results were similarly observed in studies utilising the *American College of Rheumatology (1990)* and *2016* (effect size range: −0.49 to −0.46), although only statistically significant in the *American College of Rheumatology (1990)* ($p = 0.010$; Table 4). Despite the lack of statistical significance, results were relatively similar following resistance exercise and combined resistance, aerobic and flexibility exercise (effect size range: −0.48 to −0.43).

The heterogeneity I^2 was 0%. The contour-enhanced funnel plot indicated a presence of publication bias ($\tau = -2.5$, $p = 0.030$; Supplementary Fig. S5). In the meta-regression, age, duration of the program, and inclusion of aerobic and flexibility exercise were not significantly associated with changes in depression ($p = 0.773$ to 0.991; Supplementary Table S19). A very low certainty of evidence was graded for depression based on the GRADE approach.

3.6. Fatigue

Resistance-based exercise programs resulted in a small decrease in fatigue, with an effect of −0.48 SMD (95% CI: −0.69 to −0.26, $p < 0.001$) compared to control groups. The main analysis included 18 effect sizes from 14 studies, encompassing a total of 964 patients (Supplementary Fig. S6). An improvement of 10.76 points (95% CI: 3.86

Table 3
Overall and subgroup analyses of resistance-based exercise effects on quality of life in patients with fibromyalgia.

Analysis	Descriptive			Random-effect meta-analysis		Heterogeneity			Certainty of evidence
	k	ES	n	Effect size (95% CI)	p-value	Q	I^2	p-value	
Overall, SMD	13	13	564	0.28 (0.09 to 0.46)	0.010	4.6	0%	0.971	⊖⊖⊖⊖ Very low[a,b,c,d]
Questionnaire									
SF-36 – General health, points	9	9	426	3.04 (0.57 to 5.52)	0.020	15.2	47%	0.056	
Fibromyalgia criteria									
1990, SMD	12	12	536	0.26 (0.07 to 0.46)	0.010	4.5	0%	0.954	
2016, SMD	1	1	28	0.47 (−0.29 to 1.22)	-	-	-	-	
Exercise modality									
Resistance, SMD	7	7	336	0.29 (−0.06 to 0.64)	0.090	3.8	0%	0.698	
Resistance + Flexibility, SMD	1	1	28	0.47 (−0.29 to 1.22)	-	-	-	-	
Resistance + Aerobic + Flexibility, SMD	5	5	200	0.23 (−0.05 to 0.51)	0.080	0.6	0%	0.963	

95% CI, 95% confidence intervals; ES, number of effect sizes; I^2, percentage of variation across studies that is due to heterogeneity; k, number of studies; n, number of participants; Q, Cochran's Q test of heterogeneity.
[a] Certainty of evidence downgraded due to risk of bias, with studies most studies presenting with high risk in the risk of bias assessment.
[b] Certainty of evidence downgraded due to indirectness, with different results observed across the interventions and assessment methods examined.
[c] Certainty of evidence downgraded due to publication bias, with visual assessment of comparison-adjusted funnel plots suggesting evidence of small-study effects.
[d] Certainty of evidence downgraded due to imprecision, with confidence intervals from interventions crossing null values.

Table 4
Overall and subgroup analyses of resistance-based exercise effects on depression in patients with fibromyalgia.

Analysis	Descriptive			Random-effect meta-analysis		Heterogeneity			Certainty of evidence
	k	ES	n	Effect size (95% CI)	p-value	Q	I^2	p-value	
Overall, SMD	11	12	560	−0.46 (−0.70 to −0.22)	<0.001	5.3	0%	0.916	⊖⊖⊖⊕ Very low[a,b,c]
Questionnaire									
Beck Depression Scale, points	9	9	395	−4.58 (−8.08 to −1.08)	0.020	48.6	49%	<0.001	
Fibromyalgia criteria									
1990, SMD	9	10	500	−0.46 (−0.74 to −0.17)	0.010	5.2	0%	0.812	
2016, SMD	2	2	60	−0.49 (−1.60 to 0.61)	0.110	0.0	0%	0.840	
Exercise modality									
Resistance, SMD	5	5	322	−0.43 (−0.97 to 0.10)	0.080	3.0	0%	0.558	
Resistance + Flexibility, SMD	1	1	28	−0.40 (−1.15 to 0.35)	-	-	-	-	
Resistance + Aerobic, SMD	1	1	32	−0.58 (−1.29 to 0.13)	-	-	-	-	
Resistance + Aerobic + Flexibility, SMD	4	5	178	−0.48 (−1.14 to 0.18)	0.090	2.2	0%	0.694	

95% CI, 95% confidence intervals; ES, number of effect sizes; I^2, percentage of variation across studies that is due to heterogeneity; k, number of studies; n, number of participants; Q, Cochran's Q test of heterogeneity.
[a] Certainty of evidence downgraded due to risk of bias, with studies most studies presenting with high risk in the risk of bias assessment.
[b] Certainty of evidence downgraded due to indirectness, with different results observed across the interventions and assessment methods examined.
[c] Certainty of evidence downgraded due to publication bias, with visual assessment of comparison-adjusted funnel plots suggesting evidence of small-study effects.

to 17.65, p = 0.010) on the *SF-36 vitality scale* and a reduction of −1.56 points (95% CI: −3.83 to 0.72, p = 0.100) in the fatigue domain of the Fibromyalgia impact questionnaire were observed. The primary effect was notably seen in studies utilising the *American College of Rheumatology (1990)* (−0.48 SMD, 95% CI: −0.73 to −0.23, p < 0.001; Table 5). Although not statistically significant, combined resistance, aerobic and flexibility exercise resulted in a moderate effect of 0.58 SMD (95% CI: −1.32 to 0.16, p = 0.080), while resistance exercise alone had an effect of −0.41 SMD (95% CI: −0.80 to −0.01, p = 0.050).

The heterogeneity I^2 was 0%. The contour-enhanced funnel plot indicated a presence of publication bias (τ = −5.1, p < 0.001; Supplementary Fig. S7). Meta-regression analyses indicated that age, BMI and overall symptomatology were not associated with changes in fatigue (p = 0.372 to 0.895). Characteristics of resistance exercise prescription, such as duration and weekly volume, as well as the inclusion of aerobic and flexibility exercise, were not significantly associated with changes in quality of life (p = 0.295 to 0.574). A very low certainty of evidence was graded for fatigue based on the GRADE approach.

3.7. Anxiety, sleep quality, physical function and muscle strength

A total of 5 to 6 studies, involving between 228 and 345 patients, were included in the main analysis for anxiety, sleep quality, physical function and muscle strength outcomes. Small improvements were observed for anxiety (−0.31 SMD, 95% CI: −0.43 to −0.19, p < 0.001) and sleep quality (0.39 SMD, 95% CI: 0.11 to 0.66, p = 0.020) following resistance-based exercise programs compared to control groups. A large effect was observed for lower-limb muscle strength (1.17 SMD, 95% CI: 0.89 to 1.44, p < 0.001). Regarding physical function assessed by the 6-min walking test, resistance-based exercise programs led to an improvement of 0.46 SMD (95% CI: 0.10 to 0.82, p = 0.020) and an increased distance of 29.0 m (95% CI: 10.99 to 46.97, p = 0.010) compared to control groups. Overall, the observed results were primarily derived from studies utilising the *American College of Rheumatology (1990)* and prescribing resistance exercise. Results are provided in Supplementary Table S21–S24. Due to the small number of studies included (<10), a contour-enhanced funnel plot and meta-regression were not performed. A very low certainty of evidence was graded for anxiety, sleep quality and physical function, while a low certainty of evidence was graded for lower-limb muscle strength.

4. Discussion

The present study aimed to systematically review and analyse the effects of resistance exercise programs on a range of patient-reported outcomes, including pain, quality of life, fatigue, depression, anxiety, and sleep quality, as well as objectively measured physical function and muscle strength in patients with fibromyalgia. We found that resistance-based exercise programs induced positive changes across all outcomes; however, these were relatively small and uncertain based on the GRADE approach. Nonetheless, benefits in pain, quality of life, fatigue and depression were irrespective of clinical and demographic characteristics as well as the exercise prescription. Therefore, we suggest that a low resistance exercise dosage at moderate intensity may be sufficient to achieve benefits in patients irrespective of age, BMI, duration of illness and overall symptomatology, within a multidisciplinary program targeting fibromyalgia.

Pain is the main hallmark symptom of patients with fibromyalgia (Neumann and Buskila, 2003; Queiroz, 2013). We observed that resistance exercise may play a role in managing pain and appears to improve quality of life, mainly for those diagnosed according to the *American College of Rheumatology (1990)* criteria (i.e., widespread pain in the four quadrants of the body and presence of pain on pressure in at least 11 out of 18 tender points) and irrespective of demographic and clinical characteristics. However, it should be noted that resistance exercise, as a single therapy, only resulted in improvements of <0.5 SMD (i.e., a small effect size) in pain and quality of life, in contrast to previous meta-analysis (Couto et al., 2022; Han et al., 2024; Niu et al., 2024; Rodríguez-Almagro et al., 2023). While there is no established minimal clinically important difference (MCID) for VAS pain and SF-36 bodily pain in patients with fibromyalgia, in drawing from other patients with chronic pain, we observed that the effects were below the MCID of 18.0 points for VAS (Hägg et al., 2003; Mease et al., 2011) and 11.1 points for SF-36 bodily pain (Clement et al., 2022). We hypothesised that this modest effect is likely due to lower baseline pain levels, possibly influenced using pain medications (Aster et al., 2022) or even other interventions part of usual care, despite the lack of reporting in the studies included. It is estimated that one in every three to four fibromyalgia patients takes pain medication (Aster et al., 2022), such as anti-inflammatories, analgesics, or muscle relaxants, which may have contributed to the relatively small baseline pain levels (visual analogue scale, median of 54 out of 100, IQR: 46.5-61.9), and consequently less room for further improvement through resistance exercise. Thus, while resistance exercise alone may not fully address pain in patients with fibromyalgia, it may represent a crucial component of a comprehensive, multidisciplinary therapy program (Macfarlane et al., 2017; Sarzi-Puttini et al., 2011), particularly for those with higher care needs.

With the increasing use of multicomponent exercise interventions in

Table 5
Overall and subgroup analyses of resistance-based exercise effects on fatigue in patients with fibromyalgia.

Analysis	Descriptive			Random-effect meta-analysis		Heterogeneity			Certainty of evidence
	k	ES	n	Effect size (95% CI)	p-value	Q	I^2	p-value	
Overall, SMD	14	18	964	−0.48 (−0.69 to −0.26)	<0.001	7.3	0%	0.980	⊖⊖⊖⊕ Very low[a,b,c]
Questionnaire									
SF-36 – Vitality, points	9	9	403	10.76 (3.86 to 17.65)	0.010	106.8	50%	<0.001	
Fibromyalgia impact questionnaire – Fatigue, points	4	5	283	−1.56 (−3.83 to 0.72)	0.100	4.5	64%	0.340	
Fibromyalgia criteria									
1990, SMD	12	16	606	−0.48 (−0.73 to −0.23)	<0.001	7.2	0%	0.952	
2010, SMD	1	1	41	−0.38 (−1.00 to 0.24)	-	-	-	-	
2016, SMD	1	1	28	−0.59 (−1.35 to 0.17)	-	-	-	-	
Exercise modality									
Resistance, SMD	7	9	406	−0.41 (−0.80 to −0.01)	0.050	4.0	0%	0.860	
Resistance + Aerobic, SMD	1	1	41	−0.38 (−1.00 to 0.24)	-	-	-	-	
Resistance + Flexibility, SMD	1	1	28	−0.59 (−1.35 to 0.17)	-	-	-	-	
Resistance + Aerobic + Flexibility, SMD	5	7	200	−0.58 (−1.32 to 0.16)	0.080	2.8	0%	0.831	

95% CI, 95% confidence intervals; ES, number of effect sizes; I^2, percentage of variation across studies that is due to heterogeneity; k, number of studies; n, number of participants; Q, Cochran's Q test of heterogeneity.
[a] Certainty of evidence downgraded due to risk of bias, with studies most studies presenting with high risk in the risk of bias assessment.
[b] Certainty of evidence downgraded due to indirectness, with different results observed across the interventions and assessment methods examined.
[c] Certainty of evidence downgraded due to publication bias, with visual assessment of comparison-adjusted funnel plots suggesting evidence of small-study effects.

clinical practice, programs are often designed to target broader aspects of health in patients with fibromyalgia. Interestingly, despite adding aerobic or flexibility exercises, we did not observe any additional reduction in pain or improvement in quality of life, as well as increasing the amount of weekly exercise and intensity in the meta-regression analyses. One potential explanation for this finding is the fact that most patients had no experience in resistance exercise prior the intervention, presenting with a lower threshold for adaptations, which lead to similar adaptations within the different dosages and intensities prescribed (Radaelli et al., 2024). Nonetheless, such information is important for a better planning of multidisciplinary management (Macfarlane et al., 2017). A low-dose, moderate-intensity resistance exercise program (e.g., twice a week, five exercises, two sets per exercise at 60% of 1-RM), which requires less time and effort, was sufficient to yield the observed benefits while building on compliance and maintaining adherence in this group (Vancampfort et al., 2024). Given the demanding treatment regimen and regular appointments, a low resistance exercise dosage at moderate intensity can be effectively combined with psychological support, nutrition counselling, and alternative therapies, without imposing a significant time burden on patients (Macfarlane et al., 2017). Thereafter, a higher dosage of resistance exercise may be gradually implemented to induce further benefits according to the exercise principles of *progression* and *diminishing returns*. Therefore, this multidisciplinary approach, which addresses both physical and psychological aspects of the disease (Sarzi-Puttini et al., 2011), may be a balanced and effective strategy for managing fibromyalgia symptoms. Although we offer some strategies, further studies are needed to identify the most effective combinations of therapies using patient-centred approaches.

Depression and anxiety are common comorbidities that can influence the severity of pain perception and disease in patients with fibromyalgia (Gracely et al., 2012; Yepez et al., 2022). Five out of the nine analysed studies had average scores above the threshold for moderate-to-severe depression (≥20). Despite the small effect of <0.5 SMD, a reduction of −4.6 points was observed in the Beck Depression Inventory (BDI) following resistance-based exercise programs. This effect approached the MCID of −5.0 points for BDI (Masson and Tejani, 2013). In addition, we observed that resistance-based exercise programs helped reduce depression scores, bringing patients closer to a range that is below the threshold for severe depression. As observed in the meta-regression, a low resistance exercise dosage at moderate intensity was sufficient to reduce depression levels. On the other hand, we observed a modest effect of −0.3 SMD in anxiety. The low baseline levels in the different questionnaires assessed suggest that anxiety symptoms were perhaps less pronounced, which may explain the modest effect of the intervention. The few studies included (n = 5) also precluded further analysis of resistance exercise dosage and intensity. Although a meaningful effect was not observed, resistance-based exercise programs may still be important to prevent further worsening of anxiety symptoms throughout the disease course. These benefits in reducing psychological distress are crucial for improving patients' pain perception and overall management of fibromyalgia symptoms - challenges often reported by those with higher levels of depression and anxiety (Gracely et al., 2012; Qureshi et al., 2021; Yepez et al., 2022).

In regard to fatigue and sleep disturbance, these ultimately impact daily functioning, physical activities, social interactions, ability to work, and consequently the quality of life of patients with fibromyalgia (Bigatti et al., 2008; Vincent et al., 2013). We found that resistance-based exercise programs can improve SF-36 vitality by 10.8 points and reduce FIQ Fatigue by −1.6 points, resulting in a reduction of 0.5 SMD in fatigue. To the best of our knowledge, there is no established MCID for these domains. We also observed improvements in sleep quality (0.39 SMD) as well as in physical function (6-min walking test, 0.46 SMD or 29.0 m) and lower-limb muscle strength (1.2 SMD). Although the effects were modest, they may be clinically relevant. The fatigue experienced by patients with fibromyalgia is often severe, chronic, and not alleviated by rest or sleep (Bigatti et al., 2008; Vincent et al., 2013). In addition, it impacts the ability to engage in activities of daily living and exercise, creating a cycle where fatigue and other symptoms feed into one another, further exacerbating deconditioning and pain. Interestingly, the lack of an association between resistance exercise dosage and changes in fatigue suggests that even a lower dosage of resistance exercise can be effectively delivered to counteract fatigue irrespective of age, BMI, duration of illness and overall symptomatology, potentially enhancing sleep quality and physical function. Therefore, resistance exercise is a key component of this comprehensive and multidisciplinary therapy program, critical for managing the multiple symptoms of fibromyalgia (Macfarlane et al., 2017).

The strengths of the present study are as follows: (1) the inclusion of 22 randomised controlled trials with 1235 patients, (2) a rigorous three-level mixed-effects meta-analysis to account for the nested structure of the effect sizes calculated from the studies included and (3) subgroup and meta-regression analyses on a range of demographic, clinical and exercise prescription characteristics. However, there are limitations worthy of comment. First, most studies had a high risk of bias, with concerns regarding the randomisation process, measurement of outcomes and selection of reported results. In addition, we also identified publication bias in three out of four outcomes assessed and high risk of indirectness. Altogether, these resulted in a low to very low certainty of evidence on the benefits of resistance exercise to this group. Second, the studies were undertaken exclusively with females and those patients diagnosed by the *American College of Rheumatology (1990)*, and, therefore, our findings cannot be generalised to males with fibromyalgia neither those diagnosed by more recent fibromyalgia criteria. More studies are required to examine the effect of resistance exercise programs in these settings. Finally, the limited evidence on anxiety, sleep quality, physical function and muscle strength precluded us to investigate potential demographic, clinic and exercise prescription moderators.

In conclusion, our findings suggest that resistance-based exercise programs should be strongly considered within a multidisciplinary approach targeting multiple symptoms in women with fibromyalgia. Given its promising but small effect (<0.5 SMD) as a single therapy, a low-dose, moderate-intensity resistance exercise program (e.g., twice a week, five exercises, two sets per exercise at 60% of 1-RM) may be a useful tool to help patients cope with symptoms while undergoing other interventions targeting other components of the disease. This approach may represent an effective cornerstone of treatment for fibromyalgia when combined with other therapies.

Language editing statement

We certify that the manuscript entitled "**A systematic review and meta-analysis on the benefits of resistance exercise in patients with fibromyalgia**" has been carefully reviewed and edited for English language quality, including grammar, clarity, and scientific writing style.

The language revision was conducted within an academic environment at an Australian institution, ensuring that the manuscript meets the standards of English required for publication in an international peer-reviewed journal.

All authors have approved the final version of the manuscript.

Patient consent statement

Patient consent was not required for this study as it is a systematic review and meta-analysis based exclusively on previously published data. No individual participant data were collected, accessed, or used directly by the authors.

All data included in this study were obtained from publicly available sources, and all original studies are appropriately cited.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to check grammar. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The results of the study are presented clearly, honestly, without fabrication, falsification, or inappropriate data manipulation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbmt.2026.05.010.

Data availability

Data used in the present study such as data extraction templates, forms and analysis will be made available upon reasonable request to the corresponding author.

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